

FORESIGHT

AI-Powered Demand & Inventory Intelligence Platform

Domain: Data Science & Analytics **Author:** Venkatesh M **Date:** September 2026

1. Abstract

FORESIGHT is an analytics prototype that helps businesses plan stock using historical SKU-level sales. It combines demand forecasting, inventory risk identification, and replenishment calculations in an interactive dashboard. The app accepts sample data or an uploaded CSV, generates forecasts, and presents inventory alerts and suggested replenishment quantities.

2. Problem Statement

Retailers may hold excess inventory or experience stockouts that cause lost sales. Spreadsheet-based planning can be difficult to scale across products and changing demand. FORESIGHT demonstrates how historical sales and machine learning can support more consistent planning.

3. Objectives

Explore historical sales; forecast near-term demand for each SKU; identify stockout and reorder risks; estimate suggested order quantities and inventory value; and provide downloadable results.

4. Technology Stack

Python, Pandas, NumPy, Scikit-learn Random Forest regression, Plotly, Streamlit, and CSV input/output.

5. Architecture & Methodology

Data is loaded from the bundled sample CSV or an uploaded file. The application validates fields, parses dates, removes invalid rows, and aggregates daily sales by SKU. Feature engineering uses lagged sales (1, 7, 14 days), rolling means (7 and 14 days), and calendar fields. A Random Forest model is used when at least 28 days of history exist; shorter histories use a moving-average baseline.

6. Inventory Calculations

Average daily forecast = forecast units divided by horizon. Reorder point = average daily forecast × (supplier lead time + safety coverage). Suggested order quantity = $\max(0, \text{ceiling}(\text{reorder point} - \text{current stock}))$. Inventory value = current stock × unit cost. These are decision-support estimates and should be reviewed before purchasing.

7. Dashboard Modules

Overview: KPIs, sales trend, inventory health, and priority watchlist. Demand Forecast: actual versus predicted sales for a selected SKU. Inventory Recommendations: risk-filtered reorder table and CSV download. Data Explorer: cleaned records and quality summary.

8. Data Description

The included synthetic dataset contains 120 days of daily sales for eight SKUs. Fields include date, SKU, category, sales, stock, lead time, and unit cost. Synthetic data is for demonstration only and should not be treated as evidence of real business performance.

9. Execution Instructions

Install Python 3.10 or newer, run `pip install -r requirements.txt`, then run `streamlit run app.py` from the project directory. Open the local URL printed in the terminal, normally `http://localhost:8501`. See README.md for Windows and macOS/Linux commands.

10. Evaluation Plan

For a stronger evaluation, use a chronological train/test split per SKU and compare the model with a naïve baseline using MAE and WAPE. Test missing values, duplicate dates, short history, zero demand, new SKUs, and promotional spikes. This prototype does not claim production-grade accuracy.

11. Limitations & Future Scope

The sample data is synthetic; the model uses a relatively simple feature set and recursive predictions. Future work includes backtesting and model comparison, confidence intervals, promotions and holidays, stockout-aware demand estimation, supplier constraints, ERP integration, authentication, and scheduled refreshes.

12. Conclusion

FORESIGHT demonstrates an end-to-end data science workflow covering data preparation, visualization, machine-learning forecasting, inventory risk classification, and actionable recommendations. A production version should use authorized real data, formal validation, and business-specific replenishment constraints.

Execution command: `streamlit run app.py`